MATH3023 Advanced Mathematics Applications

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Preface

These notes follow the course *Advanced Mathematics Applications*, taught by Lyle Noakes at the University of Western Australia in Semester 2 of 2026. The unit teaches advanced techniques of Calculus, including multivariable calculus, Green's, Stoke's and divergence theorem, complex analysis, ordinary and partial differential equations, and Sturm-Liouville theory.

0.1 More detailed overview of the subject.

These notes are about something but with more detail.

0.2 Table of Symbols

Symbol	Meaning
$\forall$	For every *or* for all
$\exists$	There exists
$\in$	In *or* is an element of
$\subset$	Is a proper subset of
$\subseteq$	Is a subset of
$\cup$	Union with
$\cap$	Intersection of
$\setminus$	Not *or* without
$\varnothing$	Empty set
$\sim$	Relates to
$>$	Greater than
$<$	Less than
$\geq$	Greater than or equal to
$\leq$	Less than or equal to
$\lvert n \rvert$	The absolute value of n
$\square$	It has been proven

Chapter 1

Review of Multivariable Calculus

1.1 Lecture 1

1.1.1 Double and Triple Integrals

NEW THOUGHT about the lecture